DAY 1: INTRODUCTION TO NETWORK PROTOCOLS

Introduction

Network protocols are rules that allow devices and applications to communicate. DNS helps a device find the IP address for a website name. HTTP is used by a browser and a web server to request and transfer web resources. HTTPS protects HTTP communication using TLS encryption and server authentication.

Task 1: Observe DNS and HTTP/HTTPS Traffic in Wireshark

Aim: To capture traffic while opening www.spotify.com and identify at least two DNS packets and two HTTP/HTTPS packets.

Steps:

1. Open Wireshark and select the active Wi-Fi or Ethernet interface.
2. Start the capture, then open a browser and visit https://www.spotify.com.
3. Stop the capture after the page has loaded.
4. Use the display filter dns to find DNS packets. Record two queries or responses from the actual capture.
5. For web traffic, try tls or tcp.port == 443 for HTTPS connections. Use http for unencrypted HTTP packets if any are present.
6. Write the packet details from your own capture in the tables below and add a screenshot.

Note: Spotify normally uses HTTPS, so the capture may show TLS and TCP port 443 rather than readable HTTP requests. If DNS is cached, you may not see a new DNS query. Record only packets present in your capture.

Task 2: Perform a DNS Lookup Using nslookup

Aim: To resolve www.flipkart.com to an IP address.

Command Prompt / Terminal command:

nslookup www.flipkart.com

Steps:

7. Open Command Prompt or Terminal.
8. Enter nslookup www.flipkart.com and press Enter.
9. Record the DNS server shown and the returned IP address or addresses. The site may return more than one IP address.

Task 3: Three Differences Between HTTP and HTTPS

Feature	HTTP	HTTPS
Security	Data is not encrypted by HTTP itself.	Uses TLS to encrypt traffic and help

		authenticate the server.
Default port	Usually TCP port 80.	Usually TCP port 443.
Browser address	URL begins with http://.	URL begins with https:// and browsers usually show a secure-connection indicator.

Real-world example: Instagram uses HTTPS when I log in, browse posts, or send requests. HTTPS helps protect data while it travels between the browser/app and the service.

Task 4: Observe DNS Query and Response Sequence

Aim: To observe DNS traffic for a website that has not been visited recently.

Steps:

10. Start a new Wireshark capture on the active network interface.
11. Set the display filter to dns.
12. Open a new website that you have not visited recently, for example www.mozilla.org.
13. Check the Info column for Standard query and Standard query response. Record the actual query type and response.

Typical sequence:

14. The computer sends a DNS query asking for the website's IP address, commonly an A record for IPv4 and/or AAAA for IPv6.
15. The DNS resolver sends a response containing the IP address information, or a message explaining that no answer was found.
16. The browser can then try to connect to the returned server address. This later web connection is separate from the DNS exchange.

Note: A cached DNS result may mean no new DNS query appears. Try a different domain or repeat after clearing only the browser's DNS cache if your instructor permits.

Task 5: Real-World Scenario Using DHCP

Scenario: When I connect my phone to the Wi-Fi at a café and open Instagram, the phone needs network settings to communicate. The Wi-Fi router's DHCP service can automatically assign the phone an available local IP address, subnet mask, default gateway, and DNS server information. This saves me from entering these settings manually. After the phone receives the settings, DNS can help resolve Instagram's domain name, and the app can connect to Instagram's servers using HTTPS.

Conclusion

DNS helps locate websites by resolving domain names to IP addresses. HTTP carries web requests and responses, while HTTPS adds TLS protection. Wireshark helps observe network packets, and nslookup allows a user to test DNS resolution. The packet details and IP addresses in this report must be completed using the actual results from my own computer.